

FPGA – BASED INSULIN DOSAGE COMPUTATION SYSTEM FOR DIABETES MANAGEMENT

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Abstract— Diabetes management is giving people the amount of insulin when their blood glucose level change. Normally people have to calculate the insulin dose by hand or use of a computer program, which can take time and lead to mistakes. This project is about making a machine that can automatically control insulin dose using a computer chip.

The system we made uses a programming language and a computer chip to figure out how insulin a person needs. It looks at things like how old the person is, how much they weigh, what their insulin and glucose levels were before and what their glucose level is now. Then it uses all this information to decide how much insulin they need.

We put all the information into the computer chip. It gives us a signal that says how insulin the person needs. We tested the system using a computer program to make sure it works correctly. The results show that our system can figure out the insulin dose using the information we give it.

Even though this system is not ready to be used in hospitals it shows that this kind of computer chip can be used to help people with diabetes. Maybe one day we can add sensors that can measure glucose levels in time and make it easier for doctors and nurses to use the system to help their patients.

The system we made is a start. It needs more work to be used in real life. Diabetes management is a deal and our system can help make it easier for people to get the right amount of insulin. We think that our system can be used to help people with diabetes in the future. Diabetes management is about giving people the amount of insulin and our system can do that. Diabetes management is what our system is about and we hope it can make a difference, for people with diabetes.

Keywords— FPGA, Verilog HDL, Insulin Dosage, Diabetes Management, FSM, Vivado.

I. INTRODUCTION

Diabetes is a chronic medical condition that requires continuous monitoring of blood glucose levels. Proper insulin administration is necessary to maintain normal physiological functioning. Incorrect insulin dosage may result in serious health complications, making accurate insulin estimation an important part of diabetes management. In this project, an automated insulin control system is proposed using an FPGA-based implementation.

The proposed system is developed using Verilog HDL and Finite State Machine (FSM) logic to determine the required insulin dosage based on selected patient parameters. These parameters include current glucose level, previous glucose

level, previous insulin level, age and body weight. The developed model demonstrates a hardware-based approach for processing these inputs and generating an appropriate insulin output.

II. LITERATURE SURVEY

Recent developments in diabetes management have focused on automated insulin delivery systems implemented through both software and hardware approaches. Stanchieri et al. proposed an FPGA-based digital insulin-glucose regulator in which insulin calculation was carried out entirely through hardware, without relying on software. This hardware-only implementation improved speed and predictability. The system was validated using virtual patient models and demonstrated low error and low power consumption, making it suitable for real-time biomedical applications [1].

In addition to FPGA-based designs, several studies have explored closed-loop insulin delivery systems, commonly known as artificial pancreas systems. Hovorka [2] and Doyle et al. [3] discussed the development of control algorithms for monitoring. These systems improve glucose control and patient outcomes, but they often require complex algorithms and expensive medical hardware.

The application of FPGA and embedded systems in biomedical engineering has also gained attention. Pandey and Nagaria [4] presented an FPGA-based insulin pump controller that showed efficient hardware computation. Similarly, Aiolo [5] highlighted the advantages of FPGA technology in medical devices, including parallel processing capability, low power consumption and high reliability. Compared to these existing works, the proposed project focuses on a simplified FPGA-based inputs. This approach provides a low-cost and scalable model that is suitable for academic study and can be executed for future real-time applications.

III. DESIGN & METHODOLOGY

The proposed system follows a structured hardware-based method of automated insulin computations with the use of an FPGA platform. First, the system is fed with input data like current glucose level, past glucose level and patient-specific information. These inputs are handled within the FPGA module which is an assembly of numerous functional units that are accomplished to conduct synchronized and proficient calculation.

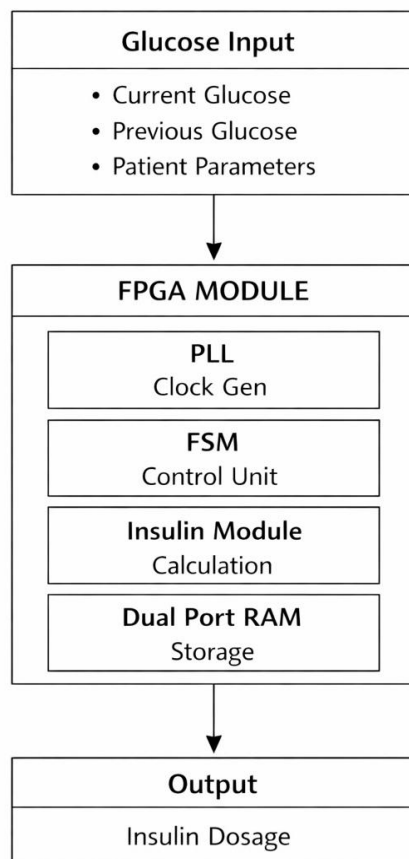


Fig. 1 Block Diagram

The FPGA board is booted up through the PLL (Phase Locked Loop) which provides the necessary clock signals that enable all the internal activities to be timed and synchronized correctly. The FSM (Finite State Machine) is the control unit which regulates the order of processes including input processing, transfer of data, and insulin calculation. Insulin Calculation Module involves the use of arithmetic calculating the amount of insulin required after inputting the parameters. At the same time, the previous glucose measurements and

insulin values are stored in the Dual Port RAM allowing to perform effective reads and writes to continue the work of the system successfully. Lastly, the system shows the insulin dosage output after all inputs have been processed and all calculations made which is the amount of insulin required by a patient. The whole procedure is realized with the help of Verilog HDL and checked with Vivado tools to provide the correct, quick, and stable calculation on hardware basis.

The block diagram is the general structure of the insulin control system based on FPGA. It depicts the input parameter flow, i.e. glucose level and patient data flow into FPGA module where various processing units, e.g. PLL, FSM,

insulin calculation module and memory execute their functions. The last result is the calculated amount of insulin dosage.

IV. IMPLEMENTATION

1) SOFTWARE IMPLEMENTATION

The software implementation involves designing, coding and verifying the insulin control system using Vivado and Verilog HDL

a) *DESIGN ENTRY*: The design Architecture is designed based on the block diagram, including modules such as FSM, insulin calculation unit, PLL and memory.

b) *CODING IN VERILOG HDL*: The complete logic of the system is written using Verilog HDL. Individual modules are created for FSM control, arithmetic operations and data storage.

c) *SIMULATION*: The design is tested using the Vivado Simulator. Different input conditions are applied to verify correct insulin output.

d) *SYNTHESIS*: The verified design is converted into a hardware-level netlist using Vivado synthesis tools.

e) *IMPLEMENTATION*: The synthesized design is mapped onto FPGA resources such as LUTs, flip-flops and RAM.

f) *BITSTREAM GENERATION*: A bitstream (.bit file) is generated, which contains the final configuration data for the FPGA.

2) HARDWARE IMPLEMENTATION

The hardware implementation involves deploying the design onto the FPGA board and testing it.

a) *FPGA SETUP*: The Nexys A7 FPGA board is connected to the computer using a USB cable for power and programming.

b) *PROGRAMMING FOR FPGA*: The generated bitstream file is uploaded to the FPGA using Vivado.

c) *INPUT CONFIGURATION*: Input parameters such as glucose levels are provided through switches or predefined test inputs.

d) *SYSTEM EXECUTION*: The FPGA executes the Verilog design, where FSM controls the process, and the insulin calculation module computes the output.

e) *DATA STORAGE*: Dual Port RAM stores previous glucose and insulin values for continuous processing.

f) **OUTPUT DISPLAY:** The calculated insulin dosage is displayed using LEDs or 7-segment display.

g) **TESTING & VERIFICATION:** The hardware output is observed and verified to ensure correct system performance.

A. PROPOSED DESIGN

The proposed system is an insulin control system based on an automated system on FPGA that determines the amount of insulin to be used based on hardware logic. The implementation of the system is based on the Verilog HDL which is simulated in the Vivado environment and a Finite State Machine that process the input parameters i.e. current glucose level, past glucose level, patient weight, patient age and past insulin levels etc. The system then processes the input parameters and after doing arithmetic operations, an adequate insulin output signal is produced.

The proposed system is very efficient and reliable compared to the traditional software implementation in that the operations will be carried out in parallel and hence faster. This system is created as a proof of concept model which is simulated with Vivado environment and the system is tested as a result. The real-time glucose sensors, etc., can be added to the current system to make the real implementation of the system in the sphere of diabetes management.

V. RESULTS & DISCUSSION

The operation of the proposed system simulated on the FPGA platform proves that the system has the correct operation in the form of five modules: PLL, Insulin Module, FSM, RAM and control logic. There are five input parameters in the system, that is, age, weight, current glucose level, previous glucose level, and previous insulin dosage, to be computed.

The calculations of insulin dosage are involved in the insulin module and the FSM controls the functionality of the system. The clock to the system is produced by the PLL. The calculated dosage of insulin is then stored in the RAM and sent through the control logic. The results of the waveforms have revealed the correct determination of insulin dose, therefore, confirming the correctness of integration between all modules in the system.

```
weight      = 16'd70;
age         = 16'd30;
prev_insulin = 16'd10;
prev_glucose = 16'd7;
current_glucose = 16'd10;
```

Fig. 2 Input Parameters for Insulin Computation

The data that the FPGA-based system received, such as weight, age, past insulin dosage, past glucose level, and present glucose level as shown in fig. 2. These values are specified in the testbench as fixed inputs and used in the insulin module in the simulation. According to these parameters, the data is processed in the system to calculate

the necessary insulin dosage. Such arrangement gives an opportunity to check the design under certain conditions and guarantee proper working of the computation logic.

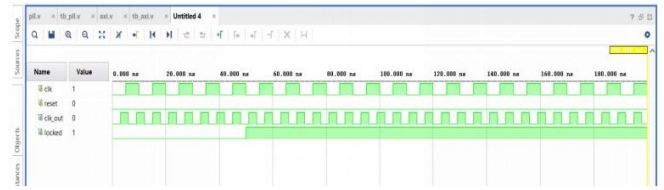


Fig. 3 Output Waveform of PLL

The simulation signal of the PLL module producing a stable clock signal as an output of the input clock. Output clock (clk out) is synchronized and locked signal is used to ensure that the system is operating smoothly. This gives good timing assurance to the entire modules in the FPGA design as shown in fig. 3.

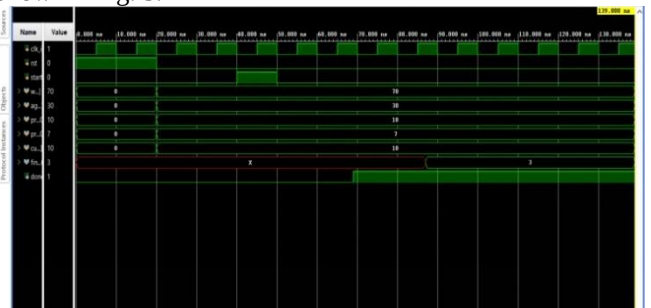


Fig. 4 Output Waveform of FSM Control

The insulin module receives the input parameters as weight, age, prev insulin, prev glucose and current glucose which are taken in the testbench. The FSM is governed by control signals such as clk, rst and start and it regulates the flow of the execution of the system. These inputs are computed to produce the final insulin output that settles down and the done signal is used to show that the process has been completed successfully as shown in Fig. 4.



Fig. 5 Output Waveform Of Dual Port RAM

The waveform shows read and write activities using signals like data_in and data_out. It retrieves stored data after initial undefined states, confirming that the memory works correctly. The RAM module serves as temporary storage for the computed insulin values. This setup allows smooth data transfer between the insulin module and other parts of the system as shown in fig. 5.

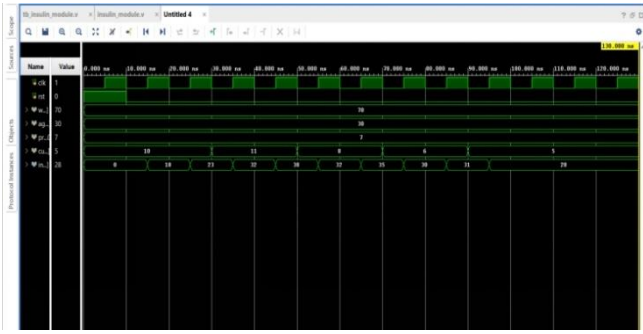


Fig. 6 Output Waveform of Insulin Computation Module

The simulation waveform of the insulin module shows what happens when we put in details like the persons age, weight and glucose levels as shown in fig. 6. We give these details to the testbench and the insulin module uses them to figure out how much insulin is needed. The waveform is like a graph that shows how the insulin_out signal changes when we change the details we put in. This helps us see that the insulin module is working correctly and giving us the amount of insulin based on the details we give it. The insulin module is doing its job. The waveform proves it by showing us how the, insulin_out signal changes when we change the age, weight and glucose levels.



Fig. 7 Output Waveform Of Top Module

The insulin module simulation shows us what the waveform looks like when we give it details like age, weight and glucose levels as shown in fig. 7. We put these details into the testbench and the insulin module uses them to figure out how much insulin is needed. The waveform shows how the insulin output changes when the inputs are different which proves that the computation logic, in the design is working correctly for the insulin module. The insulin module is really important here because it is what we are testing to make sure it is giving us the insulin dosage.

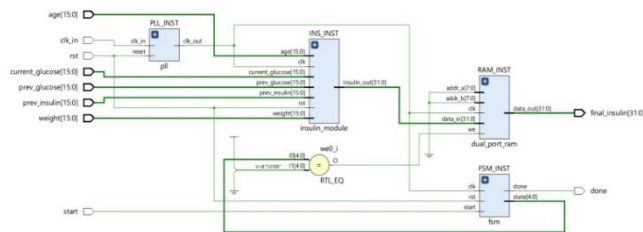


Fig. 8 Schematic Diagram

The design consists of five main modules: PLL, Insulin computation module, FSM, Dual Port RAM and control logic as shown in fig. 8. Input parameters such as age, weight, current glucose, previous glucose and previous insulin are provided to the top insulin module for processing.

```
1 set_property IOSTANDARD LVCMOS33 [get_ports {final_insulin[7]}]
2 set_property IOSTANDARD LVCMOS33 [get_ports {final_insulin[6]}]
3 set_property IOSTANDARD LVCMOS33 [get_ports {final_insulin[5]}]
4 set_property IOSTANDARD LVCMOS33 [get_ports {final_insulin[4]}]
5 set_property IOSTANDARD LVCMOS33 [get_ports {final_insulin[3]}]
6 set_property IOSTANDARD LVCMOS33 [get_ports {final_insulin[2]}]
7 set_property IOSTANDARD LVCMOS33 [get_ports {final_insulin[1]}]
8 set_property IOSTANDARD LVCMOS33 [get_ports {final_insulin[0]}]
9 set_property IOSTANDARD LVCMOS33 [get_ports {rst}]
10 set_property IOSTANDARD LVCMOS33 [get_ports {done}]
11 set_property IOSTANDARD LVCMOS33 [get_ports {clk_in}]
12 set_property PACKAGE_PIN M3 [get_ports {clk_in}]
13 set_property PACKAGE_PIN V10 [get_ports {rst}]
14 set_property PACKAGE_PIN U16 [get_ports {final_insulin[0]}]
15 set_property PACKAGE_PIN V16 [get_ports {final_insulin[1]}]
16 set_property PACKAGE_PIN T15 [get_ports {final_insulin[2]}]
17 set_property PACKAGE_PIN U14 [get_ports {final_insulin[3]}]
18 set_property PACKAGE_PIN V16 [get_ports {final_insulin[4]}]
19 set_property PACKAGE_PIN V15 [get_ports {final_insulin[5]}]
20 set_property PACKAGE_PIN V14 [get_ports {final_insulin[6]}]
21 set_property PACKAGE_PIN V12 [get_ports {final_insulin[7]}]
22 set_property PACKAGE_PIN V11 [get_ports {done}]
```

Fig. 9 FPGA Constraint File (XDC)

The XDC constraint file used for mapping input and output ports to FPGA pins as shown in fig. 9. It defines I/O standards and pin assignments for signals such as final_insulin, clk_in and done, enabling proper hardware implementation on the Nexys A7 board.

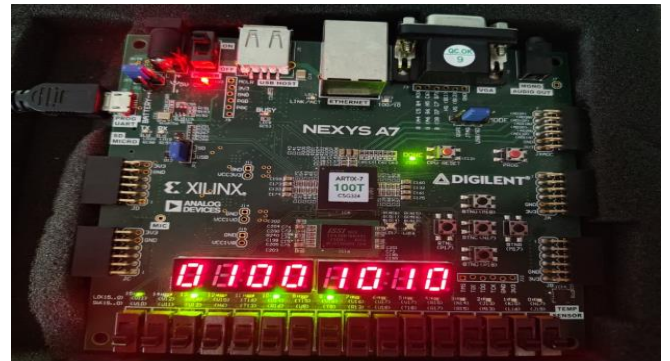


Fig. 10 FPGA Hardware Implementation

The computed insulin value is displayed using the on-board seven-segment display and LEDs as shown in fig. 10. Since the FPGA design operates using integer representation, the displayed value “63” corresponds to 6.3 IU of insulin. This confirms the successful hardware realization and output visualization of the system.

VI. CONCLUSION & FUTURE SCOPE

The project shows that an insulin dosage control system can work well using a kind of computer chip called an FPGA. This system takes in information about glucose. Produces insulin output using a specific type of logic. When we tested the system, it worked correctly which means it is an example of how this can be done with hardware.

The insulin dosage control system can be made better by adding things like real-time glucose sensors devices that convert signals and wireless communication modules. We can also make it more advanced, by using algorithms and turning it into a portable device that people can use to manage

their diabetes in real time with the insulin dosage control system.

VII. REFERENCES

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